

HOMEWORK 8

1. Let G be a group. Prove that the category $\underline{G}^{op}$ is equivalent to $\underline{G}$.
2. Let $F : \mathcal{C} \rightarrow \mathbf{Sets}$ be a functor. Consider a new category $\mathcal{D}$ with objects the pairs (X, u) , where X is an object in $\mathcal{C}$ and $u \in F(X)$. A morphism between (X, u) and (X', u') in $\mathcal{D}$ is a morphism $f : X \rightarrow X'$ in $\mathcal{C}$ such that $F(f)(u) = u'$. Prove that if (X, u) is an initial object in $\mathcal{D}$, then the functor F is represented by X .
3. Prove that the category $\mathbf{Sets}^0$ is not equivalent to $\mathbf{Sets}$.
4. Construct a functor $F : \mathbf{Groups} \rightarrow \mathbf{Sets}$ that assigns to each group the set of all its subgroups. Is this functor representable?
5. Prove that if $F : \mathcal{C} \rightarrow \mathcal{D}$ is a full and faithful functor, then $\mathcal{C}$ is equivalent to a full subcategory of $\mathcal{D}$.
6. a) Prove that if $F : \mathcal{C} \rightarrow \mathcal{D}$ is a full and faithful functor, then X and Y are isomorphic in $\mathcal{C}$ if and only if $F(X)$ and $F(Y)$ are isomorphic in $\mathcal{D}$.
b) Prove that if $F : \mathcal{C} \rightarrow \mathcal{D}$ is an equivalence of categories, then F yields a bijection between isomorphism classes of objects in $\mathcal{C}$ and $\mathcal{D}$ respectively.
7. Let $F : \mathcal{C} \rightarrow \mathcal{D}$ be a functor. Suppose that F has a left adjoint functor. Prove that F takes the terminal objects of $\mathcal{C}$ to the terminal objects of $\mathcal{D}$.
8. Construct a left adjoint functor to the functor $F : \mathbf{Groups} \rightarrow \mathbf{Sets}$ that assigns to each group G the set $\{g \in G : g^2 = e\}$.
9. Let X be a set. Prove that the represented functor $R^X : \mathbf{Sets} \rightarrow \mathbf{Sets}$ has a left adjoint functor.
10. Consider a category $\mathcal{C}$ with the set of objects $\{0, 1, 2, \dots\}$ and morphisms $Mor(i, j)$ the set of all $j \times i$ -matrices over $\mathbb{R}$ (with composition the matrix multiplication). Construct an equivalence between $\mathcal{C}$ and the category of real finite dimensional vector spaces over $\mathbb{R}$ and linear maps.